

(a) Paged Mixed-Precision KV Cache Memory

Physical GPU DRAM (GDDR7 / HBM) - 16-Token Page Blocks

Exact Attention Sinks (S4: Tokens 0-63)

FP16 (2B/elem)

Page 0 [0..15]
16 x 128 FP16

Page 1 [16..31]
16 x 128 FP16

Page 2 [32..47]
16 x 128 FP16

Page 3 [48..63]
16 x 128 FP16

Historical Quantized Pages (Q: N Pages)

INT8 (1B/elem)

Physical Page p Layout (Tokens t .. t+15, Channels d=128)

Tokens Signed INT8 Codes Z_p in [-127, 127] (d=0..127)

Scales s_p

Tok 0..3 : +14 -82 +103 -04 ... -19 +67

s_p^K = 0.042

Tok 4..7 : -41 +112 -28 +09 ... -87 -12

s_p^V = 0.038

Tok 8..11: +38 -06 +74 -95 ... +44 -62

Only 2 FP16
scalars / page

Tok 12..15: -16 -63 +04 +88 ... -09 +31

Signed INT8 Cache Codes: [-127, +127] (1 Byte/element)

Scale metadata: 4 Bytes / 2048 Bytes = 0.19% (0.125 B/token overhead)

Frozen Channel Centers: c_K, c_V in R^d (FP16 Global Vectors)

Computed once during prefill; shared across all N pages in layer

Physical Page p+1 ... Page p+N-1 (Identical Block-Strided Layout)

Recent Tail Ring Buffer (T768: 768 Tokens)

Centered FP16

Page N-48..N-33
K_E, V_E (FP16)

Page N-32..N-17
Centered via c

Recent Tail (N)
Guards Dynamic

Physical Served KV Cache Memory on RTX 5090

Mistral-7B-v0.3 at Context Length 20,480 Tokens (Batch Size B=4)

FlashInfer FP16 Baseline

2,752 MiB (100%)

PageGauge INT8 (S4/A0/T768)

1,481 MiB (53.84%)

-46.16% Memory Cut

1.86x Deployment Capacity Multiplier | Enables B=8 Serving without OOM

(b) Factorized Softmax Contraction Pipeline

Conventional Fused Low-Bit Attention (Prior Art)

ALU Bottleneck

Load Packed

Codes Z, Scales s,
Zero-Points z

In-Register ALU Dequantization

$$\mathbf{X}' = (\mathbf{Z} - \mathbf{z}) * \mathbf{s}$$

Serialized CUDA Core ALU Operations

Tensor Cores

FP16 MMA
q*K^T, P*V

Critical Hardware Bottleneck in Autoregressive Serving:

Holding scales + zero-points per element pushes register file to >160 regs/thread → Warp Stalls

PageGauge Factorized Execution (Zero In-Register Dequant)

Pure Tensor Cores

1. Direct Load INT8 Codes: Z_p^K, Z_p^V into Register Fragments

Halves memory traffic across GDDR7 bus; zero per-element offset metadata loaded

64 regs

Property 1: Key Center Softmax Shift-Invariance Cancellation

$$\alpha * \mathbf{q}^T * \mathbf{K}_p^T = \alpha * \mathbf{s}_p^K * \mathbf{q}^T * (\mathbf{Z}_p^K)^T + \delta * \mathbf{1}^T$$

$$\text{softmax}(\mathbf{z} + \delta * \mathbf{1}^T) == \text{softmax}(\mathbf{z}) \text{ [Identical Distribution]}$$

✓ Constant row offset delta cancels in softmax → ZERO ALU subtractions in registers!

2. Direct Tensor Core MMA & Hoisted Fragment Scaling

NVIDIA Tensor Cores

$$\mathbf{q} * (\mathbf{Z}_p^K)^T$$

Raw INT8 Matrix Contraction

Hoisted Fragment Scale

$$\mathbf{a}_p = \mathbf{s}_p^K * (\mathbf{q} * (\mathbf{Z}_p^K)^T)$$

Multiplied ONCE to 16-elem fragment

Value Contraction: $\mathbf{u}_p = \mathbf{s}_p^V * (\mathbf{P}_p * \mathbf{Z}_p^V)$ via Tensor Cores (MMA commutes with $\mathbf{s}_p^V$)

Hardware SM Efficiency on Blackwell (SM120)

<= 64 Regs/Thread

Max SM Occupancy
Zero Register Spills

Zero ALU Dequant

Shift Cancels in Exp
Zero ALU Stalls

Pure Tensor Cores

Hardware MMA Units
Max Compute FLOPS

(c) Associative Reduction & Global Epilogue

Historical INT8 State

Pages Q in HBM

$$\begin{aligned} m_{\text{int8}} &= \max_p(a_p) \\ l_{\text{int8}} &= \sum(\exp(a_p - m)) \\ u_{\text{int8}} &= \sum(s_p^V \exp^* Z) \end{aligned}$$

Exact Tail & Sinks

S4 + T768 in HBM

$$\begin{aligned} m_{\text{exact}} &= \max_E(a_E) \\ l_{\text{exact}} &= \sum(\exp(a_E - m)) \\ u_{\text{exact}} &= \sum(\exp^* V'_E) \end{aligned}$$

Associative Online Softmax Merger (SMEM)

Numerically Exact

$$m = \max(m_{\text{int8}}, m_{\text{exact}})$$

$$l = \exp(m_{\text{int8}} - m) * l_{\text{int8}} + \exp(m_{\text{exact}} - m) * l_{\text{exact}}$$

$$u = \exp(m_{\text{int8}} - m) * u_{\text{int8}} + \exp(m_{\text{exact}} - m) * u_{\text{exact}}$$

FlashAttention-style associative state reduction across warps

Property 2: Single Global Value Center Epilogue

$$\mathbf{o} = (\mathbf{u} / \mathbf{l}) + \mathbf{c}_V^T$$

⇒ Exactly holds because $\sum_i P_i == 1$

Restores channel center $\mathbf{c}_V$ ONCE GLOBALLY in kernel epilogue;

completely eliminates N page-level bias additions from inner loop!

Final Attention Output Vector: o in R^d

Streamed directly to Out-Projection GEMM & Feed-Forward MLP

1.67x Attention Kernel Speedup

132.20 us (PageGauge) vs. 220.75 us (FlashInfer FP16)

Evaluated on NVIDIA RTX 5090 (Blackwell SM120) with 256 MiB scrubs